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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.arima.model import ARIMA

file_path = "/content/database.csv"
df = pd.read_csv(file_path, low_memory=False)

print("✅ Columns in dataset:")
print(df.columns.tolist())
print("\n🔴 First 5 rows:")
print(df.head())

df = df[df['Year'].notnull()]
df['Year'] = df['Year'].astype(int)
df = df.sort_values('Year')

crime_by_year = df.groupby('Year').size().reset_index(name='CrimeCount')
print("\n📊 Crimes per year:")
print(crime_by_year.tail())

plt.figure(figsize=(12, 5))
plt.plot(crime_by_year['Year'], crime_by_year['CrimeCount'], marker='o', linestyle='-')
plt.title("📊 Total Crime Incidents Per Year")
plt.xlabel("Year")
plt.ylabel("Crime Count")
plt.grid(True)
plt.tight_layout()
plt.show()

ts = crime_by_year.set_index('Year')['CrimeCount']
model = ARIMA(ts, order=(2, 1, 2))
model_fit = model.fit()

forecast_steps = 5
forecast = model_fit.forecast(steps=forecast_steps)
last_year = ts.index[-1]
future_years = np.arange(last_year + 1, last_year + 1 + forecast_steps)

plt.figure(figsize=(12, 5))
plt.plot(ts, label='Historical Data')
plt.plot(future_years, forecast, label='Forecast', marker='o', linestyle='--', color='red')
plt.title("🔮 Forecasted Crime Incidents for Next 5 Years")
plt.xlabel("Year")
plt.ylabel("Crime Count")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

forecast_series = pd.Series(forecast.values, index=future_years)
print("📊 Forecasted Crime Counts (next 5 years):")
print(forecast_series)

plt.figure(figsize=(12, 5))
df['Crime Type'].value_counts().head(10).plot(kind='bar', color='skyblue')
plt.title("🔍 Top 10 Crime Types (Overall)")
plt.xlabel("Crime Type")
plt.ylabel("Number of Cases")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 5))
df['City'].value_counts().head(10).plot(kind='bar', color='salmon')
plt.title("🌆 Top 10 Cities by Total Crimes")
plt.xlabel("City")

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plt.ylabel("Crime Count")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 5))
df['Weapon'].value_counts().head(10).plot(kind='bar', color='olive')
plt.title("🔪 Crimes by Weapon Used (Top 10)")
plt.xlabel("Weapon")
plt.ylabel("Number of Cases")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()

solved_trend = df.groupby(['Year', 'Crime Solved']).size().unstack().fillna(0)
solved_trend.plot(kind='line', figsize=(12, 5))
plt.title("✅ Crimes Solved vs Unsolved Over Years")
plt.xlabel("Year")
plt.ylabel("Number of Cases")
plt.grid(True)
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 4))
df['Victim Sex'].value_counts().plot(kind='pie', autopct='%1.1f%%', startangle=90,
                                     colors=['#ff9999', '#66b3ff', '#99ff99'])
plt.title("👤 Victim Sex Distribution")
plt.ylabel("")
plt.tight_layout()
plt.show()

plt.figure(figsize=(10, 4))
df['Victim Race'].value_counts().head(10).plot(kind='bar', color='purple')
plt.title("🌈 Victim Race Distribution")
plt.xlabel("Race")
plt.ylabel("Number of Victims")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()

top_city = df['City'].value_counts().idxmax()
city_df = df[df['City'] == top_city]
city_trend = city_df.groupby('Year').size()

plt.figure(figsize=(12, 5))
plt.plot(city_trend.index, city_trend.values, marker='o', linestyle='-')
plt.title(f"📍 Yearly Crime Trend in {top_city}")
plt.xlabel("Year")
plt.ylabel("Crime Count")
plt.grid(True)
plt.tight_layout()
plt.show()

```

Columns in dataset:
 ['Record ID', 'Agency Code', 'Agency Name', 'Agency Type', 'City', 'State', 'Year', 'Month', 'Incident', 'Crime Type', 'Crime Solver']

First 5 rows:

	Record ID	Agency Code	Agency Name	Agency Type	City	State
0	1	AK00101	Anchorage	Municipal Police	Anchorage	Alaska
1	2	AK00101	Anchorage	Municipal Police	Anchorage	Alaska
2	3	AK00101	Anchorage	Municipal Police	Anchorage	Alaska
3	4	AK00101	Anchorage	Municipal Police	Anchorage	Alaska
4	5	AK00101	Anchorage	Municipal Police	Anchorage	Alaska

	Year	Month	Incident	Crime Type	Victim Ethnicity
0	1980	January	1	Murder or Manslaughter	Unknown
1	1980	March	1	Murder or Manslaughter	Unknown
2	1980	March	2	Murder or Manslaughter	Unknown
3	1980	April	1	Murder or Manslaughter	Unknown
4	1980	April	2	Murder or Manslaughter	Unknown

	Perpetrator Sex	Perpetrator Age	Perpetrator Race
0	Male	15	Native American/Alaska Native
1	Male	42	White
2	Unknown	0	Unknown
3	Male	42	White
4	Unknown	0	Unknown

	Perpetrator Ethnicity	Relationship	Weapon	Victim Count
0	Unknown	Acquaintance	Blunt Object	0
1	Unknown	Acquaintance	Strangulation	0
2	Unknown	Unknown	Unknown	0
3	Unknown	Acquaintance	Strangulation	0
4	Unknown	Unknown	Unknown	0

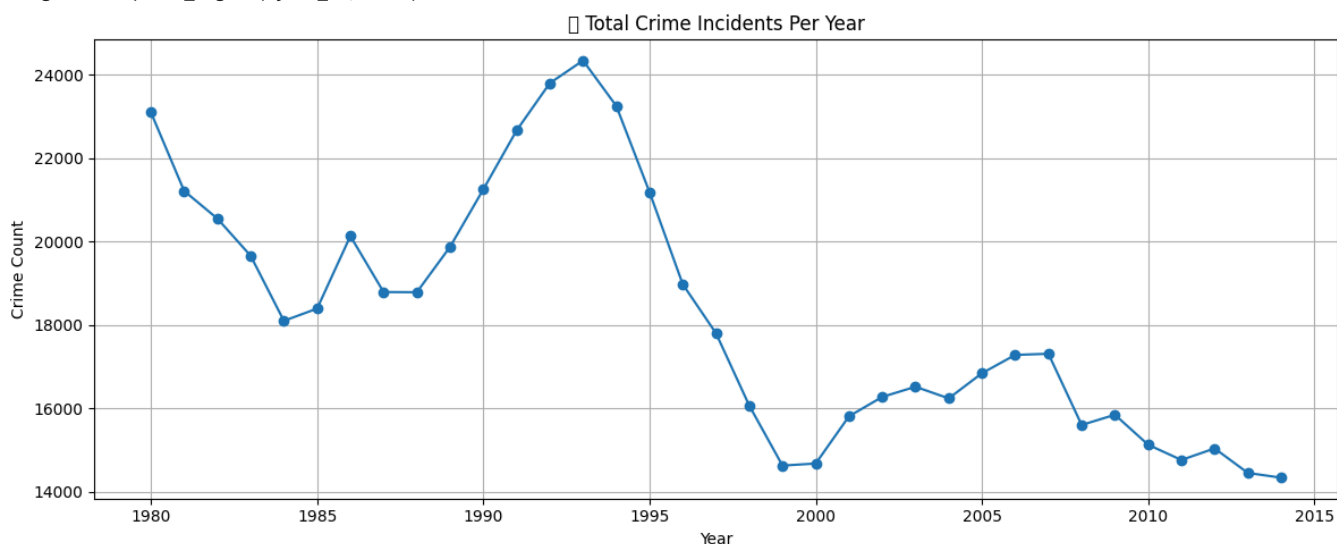
	Perpetrator Count	Record Source
0	0	FBI
1	0	FBI
2	0	FBI
3	0	FBI
4	1	FBI

[5 rows x 24 columns]

Crimes per year:

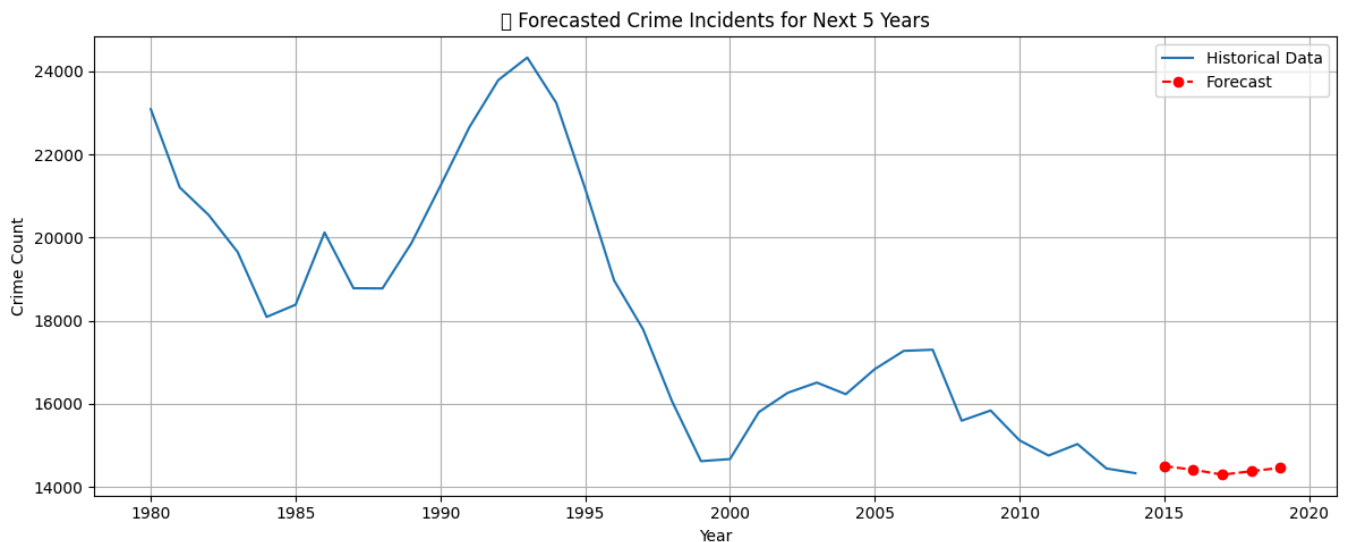
	Year	CrimeCount
30	2010	15121
31	2011	14756
32	2012	15033
33	2013	14445
34	2014	14331

```
<ipython-input-5-52e285e0ef48>:34: UserWarning: Glyph 128200 (\N{CHART WITH UPWARDS TREND}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128200 (\N{CHART WITH UPWARDS TREND}) missing from font(s) DejaVu Sans.
fig.canvas.print_figure(bytes_io, **kw)
```



```
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: An unsupported index was provided. As a result, the model will not be able to handle missing values.
self._init_dates(dates, freq)
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: An unsupported index was provided. As a result, the model will not be able to handle missing values.
self._init_dates(dates, freq)
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: An unsupported index was provided. As a result, the model will not be able to handle missing values.
self._init_dates(dates, freq)
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/statespace/sarimax.py:966: UserWarning: Non-stationary starting autoregressive parameters found. Using zero for starting values.
warn('Non-stationary starting autoregressive parameters found. Using zero for starting values.')
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameter found. Using zero for starting value.
warn('Non-invertible starting MA parameter found. Using zero for starting value.')
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:837: ValueWarning: No supported index is available. Prediction will be made using the first available index.
return get_prediction_index()
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:837: FutureWarning: No supported index is available. In the future, this will raise an exception.
return get_prediction_index()
```

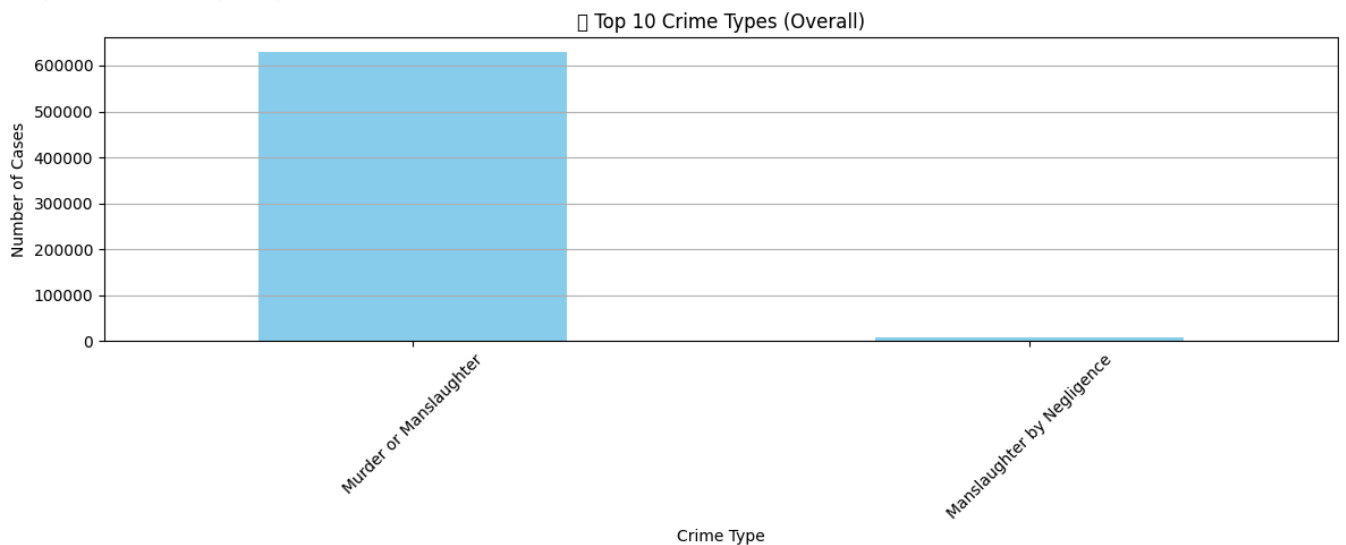
```
<ipython-input-5-52e285e0ef48>:56: UserWarning: Glyph 128302 (\N{CRYSTAL BALL}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128302 (\N{CRYSTAL BALL}) missing from font(s) DejaVu Sans.
fig.canvas.print_figure(bytes_io, **kw)
```



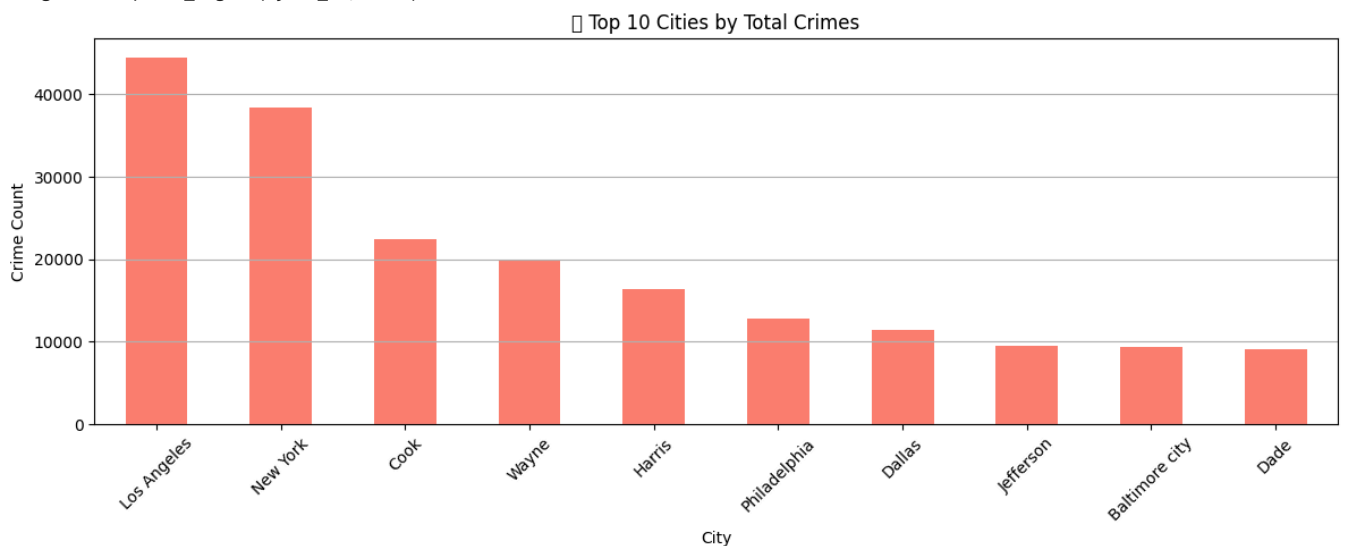
Forecasted Crime Counts (next 5 years):

```
2015 14500.374718
2016 14415.806519
2017 14295.328232
2018 14378.832252
2019 14461.178989
dtype: float64
```

```
<ipython-input-5-52e285e0ef48>:72: UserWarning: Glyph 128270 (\N{RIGHT-POINTING MAGNIFYING GLASS}) missing from font(s) DejaVu Sans
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128270 (\N{RIGHT-POINTING MAGNIFYING GLASS}) missing from font(s) DejaVu Sans.
fig.canvas.print_figure(bytes_io, **kw)
```

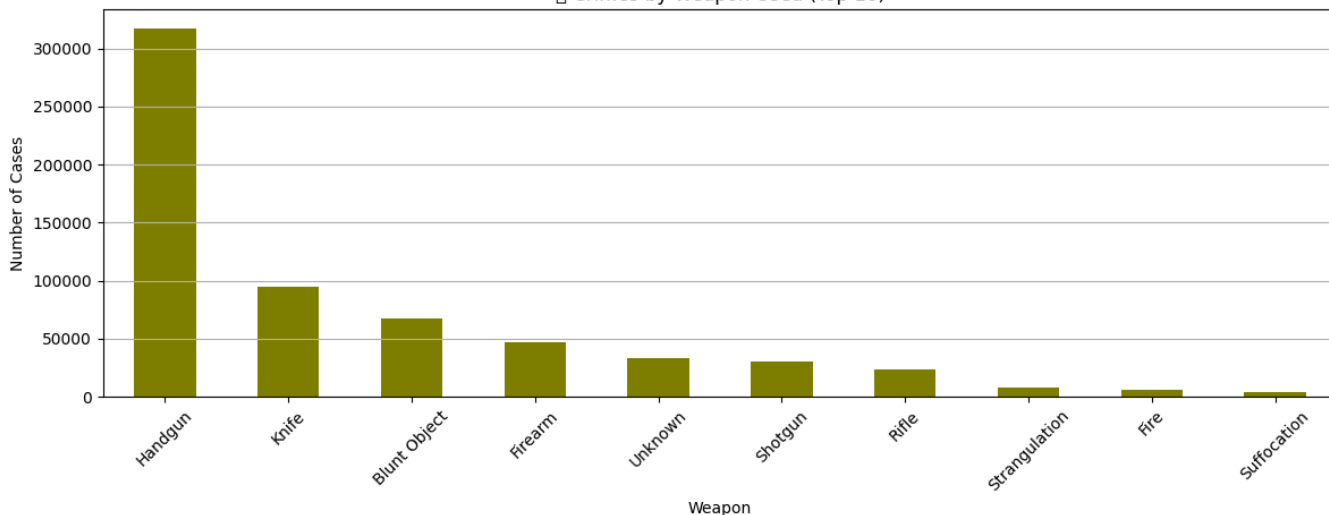


```
<ipython-input-5-52e285e0ef48>:83: UserWarning: Glyph 127961 (\N{CITYSCAPE}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 127961 (\N{CITYSCAPE}) missing from font(s) DejaVu Sans.
fig.canvas.print_figure(bytes_io, **kw)
```



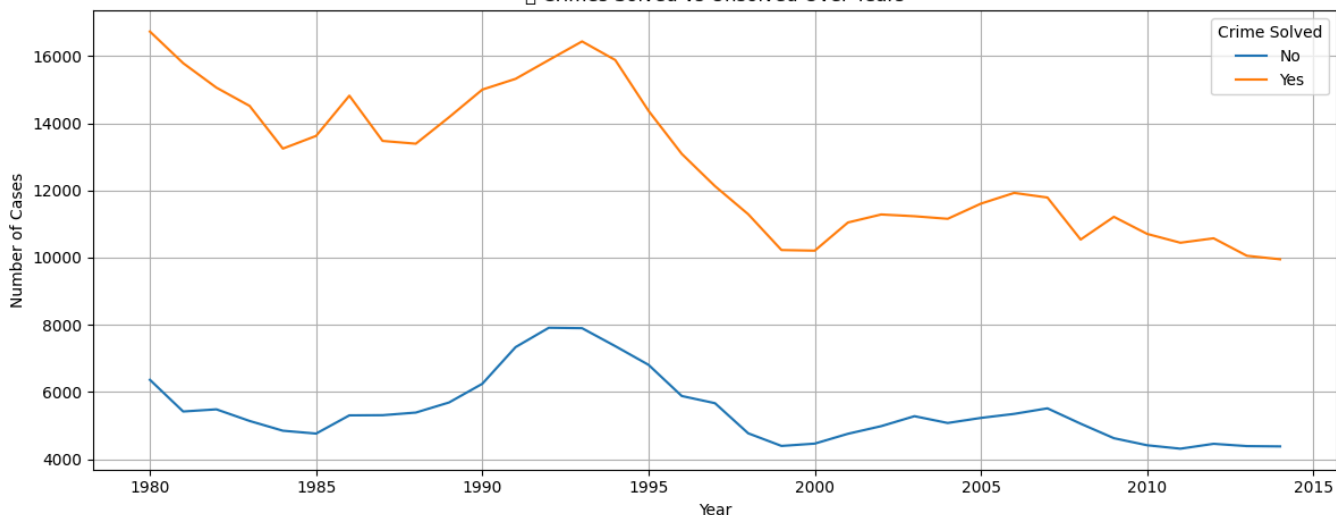
```
<ipython-input-5-52e285e0ef48>:94: UserWarning: Glyph 128299 (\N{PISTOL}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128299 (\N{PISTOL}) missing from font(s)
fig.canvas.print_figure(bytes_io, **kw)
```

Crimes by Weapon Used (Top 10)



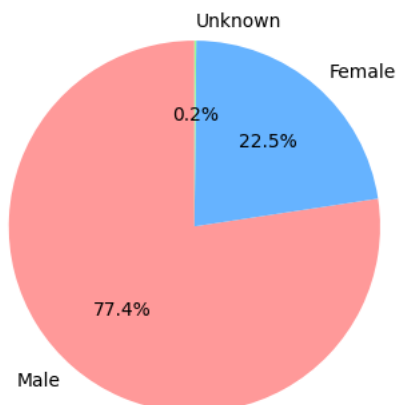
```
<ipython-input-5-52e285e0ef48>:104: UserWarning: Glyph 9989 (\N{WHITE HEAVY CHECK MARK}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 9989 (\N{WHITE HEAVY CHECK MARK}) missing
fig.canvas.print_figure(bytes_io, **kw)
```

Crimes Solved vs Unsolved Over Years



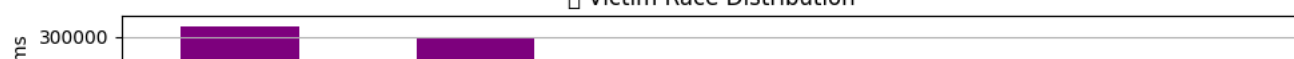
```
<ipython-input-5-52e285e0ef48>:113: UserWarning: Glyph 128101 (\N{BUSTS IN SILHOUETTE}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128101 (\N{BUSTS IN SILHOUETTE}) missing
fig.canvas.print_figure(bytes_io, **kw)
```

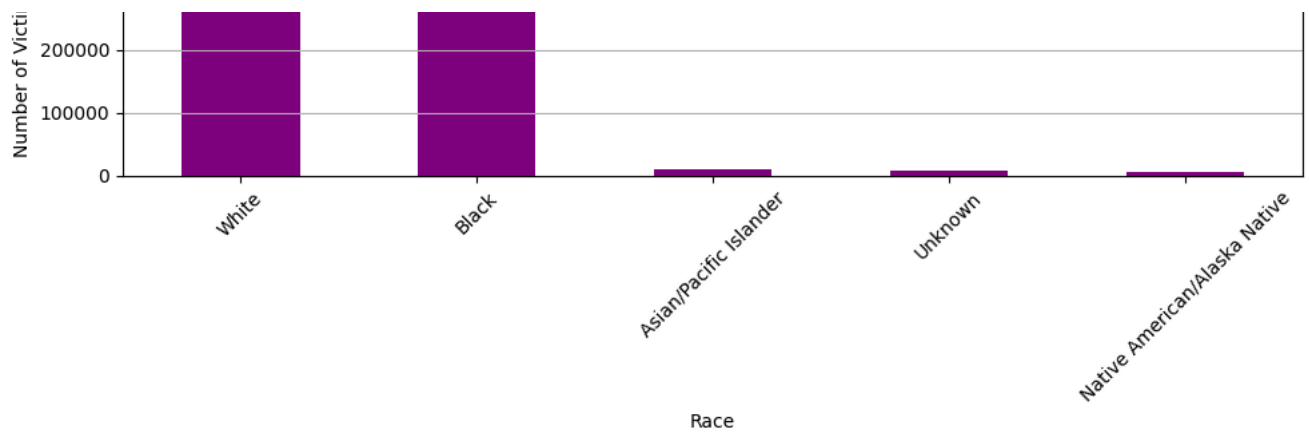
Victim Sex Distribution



```
<ipython-input-5-52e285e0ef48>:124: UserWarning: Glyph 127757 (\N{EARTH GLOBE EUROPE-AFRICA}) missing from font(s) DejaVu Sans.
plt.tight_layout()
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 127757 (\N{EARTH GLOBE EUROPE-AFRICA}) mi
fig.canvas.print_figure(bytes_io, **kw)
```

Victim Race Distribution





```
<ipython-input-5-52e285e0ef48>:138: UserWarning: Glyph 128205 (\N{ROUND PUSHPIN}) missing from font(s) DejaVu Sans.  
plt.tight_layout()  
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128205 (\N{ROUND PUSHPIN}) missing from f  
fig.canvas.print_figure(bytes_io, **kw)
```



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